Bibliography
P2P File Synchronization System

Action Learning Project

Backend (Go, C++)

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Frontend and CI/CD (Java, GitHub Actions)

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Contents

1 Identified Papers

TITLE	YEAR	CITATIONS	KEYWORDS	WHY NEEDED?	CORE?
Time, Clocks, and the Ordering of Events in a Distributed System	1978	8407	logical clocks, causality	Vector-clock baseline	Core
Kademlia: A Peer-to-Peer Information System Based on the XOR Metric	2002	2997	DHT, XOR metric	Peer discovery model	Core
The BitTorrent P2P File-Sharing System: Measurements and Analysis	2005	760	swarming, piece exchange	Chunk synchronization strategy	Yes
The rsync algorithm	1996	311	delta encoding, rolling checksum	Efficient block-level diff	Core
A comprehensive study of Convergent and Commutative Replicated Data Types	2011	256	CRDT, eventual consistency	Conflict-free merge model	Future
Brewer's conjecture and the feasibility of consistent, available, partition-tolerant web services	2002	1683	CAP, partition tolerance	Consistency trade-off baseline	Core
The Google file system	2003	5002	distributed FS, replication	Metadata and version layout	Yes
Dynamo: Amazon's Highly Available Key-Value Store	2007	382	anti-entropy, vector clocks	Replica reconciliation patterns	Yes
Paxos Made Simple	2001	977	consensus, fault tolerance	Coordination fallback option	Future
The Chubby lock service for loosely-coupled distributed systems	2006	901	leases, lock service	Leader and lock design	Yes
ZooKeeper: wait-free coordination for internet-scale systems	2010	1314	coordination, metadata nodes	Distributed coordination API	Yes
Ceph: a scalable, high-performance distributed file system	2006	1430	metadata, distributed storage	Storage architecture reference	Yes

TITLE	YEAR	CITATIONS	KEYWORDS	WHY NEEDED?	CORE?
Managing update conflicts in Bayou, a weakly connected replicated storage system	1995	813	weak consistency, anti-entropy	Conflict-policy design input	Future
inotify(7) — Linux manual page	2005	N/A	Linux kernel events	Linux file watcher API	Core
File System Events Programming Guide (Apple)	2007	N/A	macOS file events	macOS file watcher API	Core

2 Chosen Papers

TITLE	YEAR	CITATIONS	KEYWORDS	WHY NEEDED?	USED
RFC 6762: Multicast DNS	2013	N/A	mDNS, peer discovery	LAN peer discovery protocol	Phase 1
RFC 6763: DNS-Based Service Discovery	2013	N/A	DNS-SD, service advertisement	Service registration and lookup	Phase 1
Time, Clocks, and the Ordering of Events in a Distributed System	1978	8407	logical clocks, causality	Vector-clock baseline	Phase 2+
The rsync algorithm	1996	311	delta encoding, rolling checksum	File-diff algorithm for sync	Phase 2+
Brewer's conjecture and the feasibility of consistent, available, partition-tolerant web services	2002	1683	CAP, partition tolerance	Consistency trade-off baseline	Design
A comprehensive study of Convergent and Commutative Replicated Data Types	2011	256	CRDT, eventual consistency	Conflict-free merge model	Future
inotify(7) — Linux manual page	2005	N/A	Linux kernel events	Linux file watcher API	Phase 1
File System Events Programming Guide (Apple)	2007	N/A	macOS file events	macOS file watcher API	Phase 1